

Deep Multitask Learning Through Soft Layer Ordering

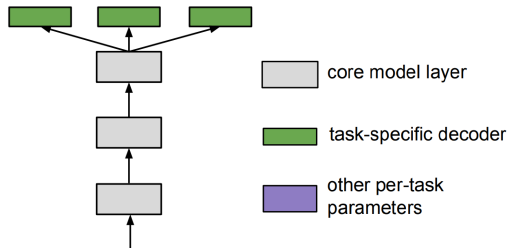
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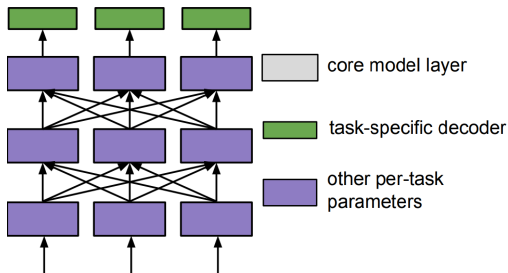
Deed MTL: classical approaches

- 1 Learning a shared representation at a high-level layer;
- 2 Task-specific decoders that extract labels for each task;
- 3 Core network is shared entirety.



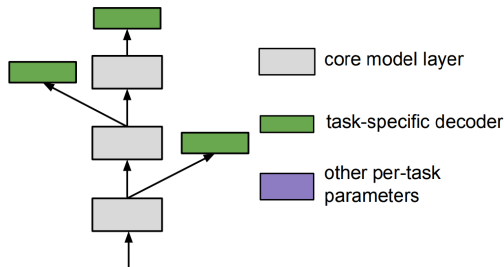
Deed MTL: column-based approaches

- 1 Assign each task its own layer of task-specific parameters at each shared depth;
- 2 Define a mechanism for sharing parameters between tasks at each shared depth.



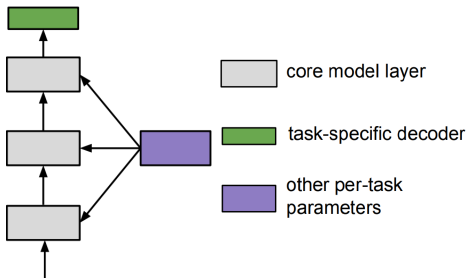
Deed MTL: supervision at custom depths

- 1 Assign each task its own layer of task-specific parameters at each shared depth;
- 2 Learning a task-relationship hierarchy during training, learned parameters are still shared across matching depths;
- 3 Include explicit recurrence that reintegrates information from earlier predictions.



Deed MTL: universal representations

- 1 Share all core model parameters except batch normalization scaling factors;
- 2 When the number of classes is equal across tasks, even output layers can be shared.



Parallel Ordering of Layers

Preliminaries:

- 1 T tasks $t_1, \dots, t_T$ are learning jointly;
- 2 t_i associated with a model $y_i = \mathcal{F}_i(x_i)$;
- 3 Sharing across tasks occurs at D consecutive depths;
- 4 $\mathcal{E}_i$ ($\mathcal{D}_i$) is t_i 's task-specific encoder (decoder);
- 5 W_k the layer of learned weights at shared depth k ;
- 6 ϕ_k an optional nonlinearity for W_k .

Parallel ordering assumption:

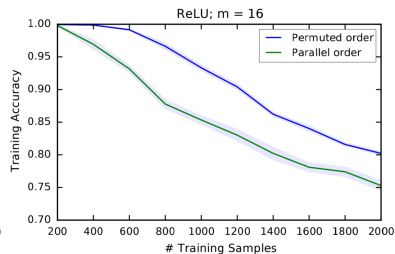
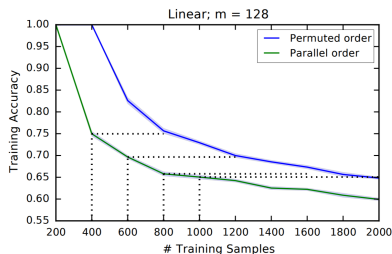
$$y_i = (\mathcal{D}_i \circ \phi_D \circ W_D \circ \phi_{D-1} \circ W_{D-1} \circ \dots \circ \phi_1 \circ W_1 \circ \mathcal{E}_i)(x_i).$$

Permuted Ordering of Layers

Permuted ordering assumption:

$$y_i = (\mathcal{D}_i \circ \phi_D \circ W_{\rho_i(D)} \circ \phi_{D-1} \circ W_{\rho_i(D-1)} \circ \dots \circ \phi_1 \circ W_{\rho_i(1)} \circ \mathcal{E}_i)(x_i),$$

where ρ_i is a task-specific permutation of size D , fixed before training.



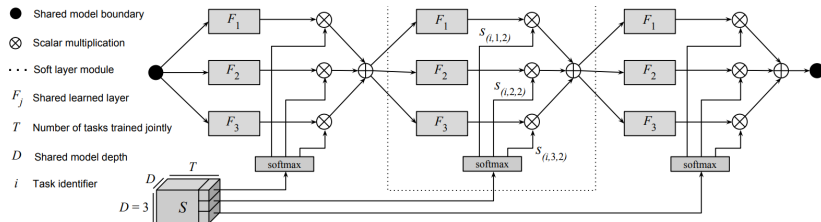
Soft Ordering of Layers

Soft ordering model:

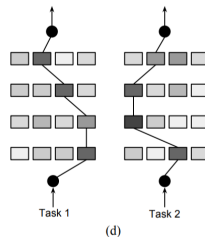
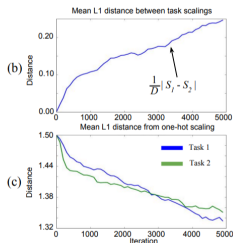
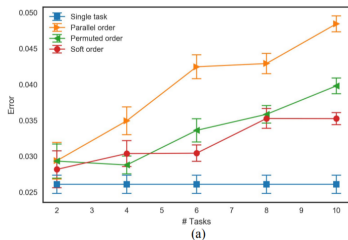
$$y_i^k = \sum_{j=1}^D s_{(i,j,k)} (\phi_k[W_j(y_i^{k-1})]), \text{ with } \sum_{j=1}^D s_{(i,j,k)} = 1 \quad \forall (i, k),$$

where $y_i^0 = \mathcal{E}_i(x_i)$, $y_i = \mathcal{D}_i(y_i^D)$;

each $s_{(i,j,k)}$ is drawn from S : a tensor of learned scales for each task t_i for each layer W_j at each depth k .



Experiments: MNIST dataset



Experiments: UCI dataset

(a)

Dataset	Input Features	Output classes	Samples
Australian credit	14	2	690
Breast cancer	30	2	569
Ecoli	7	8	336
German credit	24	2	1000
Heart disease	13	5	303
Hepatitis	19	2	155
Iris	4	3	150
Pima diabetes	8	2	768
Wine	13	3	178
Yeast	8	10	1484

(b)

